Laeeq Aslam

Energy & Urban Climate | Physics-Informed Time-Series Forecasting | Edge Deployment

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Professional Summary

PhD candidate in Control Science, developing **physics-informed AI** to advance the UN Sustainable Development Goals. My research creates **hardware-aware forecasting models** for **clean energy** (wind/load) and **urban climate resilience**, evidenced by **10+ papers (6 first-author)** and repeated SOTA gains on edge devices.

Publications & Research

Authored **10 journal** articles (**6 as first-author**) and **3 conference** papers, with **3** additional manuscripts **under review**.**Physics-Informed Ensemble Learning for City-Scale Temperature Prediction During Thermal Extremes PIEL-NET. (Urban Climate, 2025 (IF: 6.4, JCR: Q1))**.

- *Developed PIEL-NET, a hierarchical ensemble model that integrates physics-based modeling with a spatiotemporal ConvLSTM and a novel Regime Adaptive Focal Loss to predict city-core temperature by processing nine surrounding grid cells.*
- *Achieved state-of-the-art performance, reducing extreme cold prediction error by 69.2% and attaining a 0.89 Dice score for early warnings, significantly outperforming benchmarks including TCN, LSTM, and iTransformer across four global cities.*

- **Physics-Informed Spatio-Temporal Network with Trainable Adaptive Feature Selection for Short-Term Wind Speed Prediction. (Computers & Electrical Engineering, 2025 (IF: 4.9, JCR: Q1)).**
  - *Developed PISTNet, a physics-informed spatio-temporal network featuring a Dynamic Feature Adapter for automatic feature selection, a Context-Aware Attention Mechanism, and an Adaptive Physics Penalty Loss (APPL) that selectively enforces physical constraints during training.*
  - *Achieved state-of-the-art performance, outperforming eight recent models (including CoReSTL, Mixformer, and iTransformer) across four global datasets, with improvements of up to 7.1% in RMSE, 8.0% in MAE, and 9.5% in SMAPE for 6-hour wind speed prediction.*
- **A Shallow Hybrid Model with Dynamic Bayesian Optimisation for Wind Speed Prediction on Memory-Constrained Devices. (Computers & Electrical Engineering (IF: 4.9, JCR: Q1)).**
  - *Developed D-TPE-SHM, an edge-deployable shallow hybrid that combines Independent Component Analysis with adaptive Least Absolute Shrinkage and Selection Operator for location-aware feature selection, a concurrent Long Short-Term Memory and Temporal Convolutional Network temporal module, and a Randomized Vector Functional Link attention head, while a dynamic Tree-structured Parzen Estimator tunes a bounded search space for 1 to 6 hour forecasts on Jetson Nano.*
  - *Achieved a state-of-the-art efficiency–accuracy trade-off on the Mongolia dataset, reducing mean squared error by 5.9% and mean absolute error by 5.0% relative to the Ensemble Deep Randomized Vector Functional Link model, and outperforming CoReSTL and iTransformer, while shrinking parameters by 38.3% with sustained on-device latency.*
- **Dynamic Optimization of Recurrent Networks for Wind Prediction on Edge Devices. (IEEE Access, 2025 (IF: 3.6, JCR: Q2)).**
  - *Developed an adaptive simulated annealing with memory-based rejection framework to co-optimize LSTM, GRU, and TCN architectures under memory constraints, with integrated feature selection for edge-deployment on Jetson Nano.*
  - *Achieved state-of-the-art edge accuracy and efficiency on the Mongolia dataset, where the optimized Gated Recurrent Unit delivered up to 22.2% lower mean squared error and 6.3% lower mean absolute percentage error than the closest baseline Flowformer, while reducing model size by 95.7% and increasing the coefficient of determination by about 2 points.*

- **Under Review**

- *A Distribution Matching Framework for Zero-Shot Wind Speed Prediction on Edge Devices in Unseen Locations. (Under Review, Engineering Applications of Artificial Intelligence).*
- *PiDR-Net: A Perception-Informed Deep Recurrent Network for Short-Term Wind Speed Predictions (Submitted, Applied Soft Computing).*
- *Physics-Informed Extreme Wind Speed Prediction with a Multi-Gated Convolutional Recurrent Network VORTEX-Net (submitted to Applied Energy).*

Research Experience

My evolution into a **Full-Stack Researcher** began with six years of teaching embedded systems and signal processing courses at **iUSE**, establishing my theoretical foundations. This expertise enabled my transition to industry, where I designed and developed computer vision systems at **DLISION and Bond & Built**. Now at **Central South University**, my PhD research unifies these domains to propose physics-informed forecasting prediction models for sustainable systems and to validate models on edge hardware.

- **PhD Researcher – Machine Learning & AI for Sustainable Systems** | Sept 2020 – Present
  - ***Conceived and executed an independent research program*** *from problem formulation to publication, resulting in 6 first-author papers that pioneer the integration of physical constraints with deep learning for climate and energy forecasting.*
  - ***Led the full research lifecycle*** *for projects, identifying gaps in spatio-temporal forecasting, designing novel neural architectures, developing custom training paradigms (e.g., adaptive loss functions), and validating against state-of-the-art benchmarks across global datasets.*
  - *Contributed to the successful acquisition of the* ***Key R&D Program of Hunan Province (Project 2020WK2007)****, aligning doctoral research with provincial strategic goals for wind energy storage and grid optimization.*
  - *Established a rigorous experimental framework for* ***edge-AI model evaluation****, co-optimizing model architecture and hyperparameters for deployment on resource-constrained devices (NVIDIA Jetson Nano), achieving new Pareto frontiers in accuracy-efficiency trade-offs.*
  - ***Mentored and collaborated*** *with junior PhD and master's students on model development and paper writing, fostering a productive research environment.*
- **AI/ML Consultant - Edge Computing & Computer Vision** | Bond and Built Pvt Ltd, Pakistan | July 2024 – March 2025
  - *Led the* ***applied research and development*** *for a real-time footwear analytics system, tackling challenges in large-scale edge deployment and multi-stream data processing.*
  - *Scaled a computer vision pipeline to process* ***50,000+ daily inferences across 60+ concurrent streams,*** *delivering actionable market intelligence and demonstrating robust system architecture.*
- **Machine Learning Engineer – Computer Vision & AI Deployment** | DLISION, Pakistan | May 2021 – Sept 2022
  - *Led* ***computer vision projects*** *focused on* ***semantic segmentation and real-time object detection****.*
  - *Optimized* ***UNet models with*** *convolutional and attention backbones, improving* ***segmentation accuracy by 3% against their baseline architectures*** *using* ***Focal Loss*** *parameters adjustment.*
- **Higher Education Research Assistant – AI for Healthcare** | International Islamic University, Pakistan| March 2019 – Feb 2020
  - *Applied* ***Generative Adversarial Networks (GANs)*** *for* ***breast cancer detection****, improving classification accuracy.*
  - *Conducted data augmentation techniques that* ***increased dataset diversity*** *and* ***enhanced model robustness****.*
- **Lecturer & Lab Engineer – Embedded Systems & AI** |iUSE School of Engineering and Management Sciences, Pakistan| Sept 2013 – Feb 2019

- *Taking undergraduate courses related to **Embedded Systems**, **Programming,** and **Discrete Signal Processing**.*
- *Designed and developed an **Instrumentation & Measurement Lab** for **sensor-based data acquisition in MATLAB**.*
- *Led **student research projects** on **energy management, accident alerts, and secure electronic voting**.*

Education

- PhD in Control Science & Machine Learning | Central South University, China | Sept 2020 – Present.
- MS in Electronic Engineering| International Islamic University, Pakistan | Sept 2015 – Aug 2017.
- BS in Electronic Engineering| International Islamic University, Pakistan | Sept 2006 – Aug 2010.

Technical Skills

- **Machine Learning & AI:** Deep Learning | Computer Vision | Time-Series Analysis.
- **Programming & Tools:** Python | MATLAB | C++ | TensorFlow | PyTorch | OpenCV | Edge Impulse.
- **DevOps & Deployment:** Docker | AWS (EC2)| Git | Triton Inference Server.
- **Hardware & IoT:** Raspberry Pi | NVIDIA Jetson | Arduino Nano BLE Sense.
- **Data Analysis & Optimization:** Pandas | NumPy | Matplotlib | Seaborn | Feature Engineering

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Awards & Honors

- Gold Medalist – MS in Electronic Engineering | International Islamic University.
- Hunan Provincial Government Funding for Research & Development (**Project 2020WK2007**).
- Chinese Government CSC Scholarship for PhD Studies.

Professional Memberships & Certifications

- Registered Engineer, Pakistan Engineering Council (ELECTRO/22837)
- Member, Institute of Electrical and Electronics Engineers (IEEE)

References

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